

An Approach to Quantifying Placeness Using Spatial Review Text Data

Using User Reviews of Cafés in Seoul

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Placeness

Review text data

Quantification

Research Focus

**This study aims to quantify placeness,
a qualitative and subjective concept.**

Research Highlights

- Placeness has mostly been evaluated through subjective methods such as surveys and observation, which motivates a data-based and LLM-assisted quantitative approach.
- Placeness is structured into three dimensions and ten detailed factors based on placeness theory and prior studies.
- A logical scoring framework is proposed to extract, classify, and quantify placeness factors from spatial review data.
- The empirical analysis uses Google Maps reviews collected from cafés in Seoul.
- The quantified results can be extended to place recommendation, search, and comparison.

Contribution: converting qualitative place experience into comparable placeness indicators

Placeness Theory

Placeness is understood as a multidimensional concept formed through the interaction of physical environment, human activity, and meaning.

Physical characteristics

Observable spatial and environmental qualities

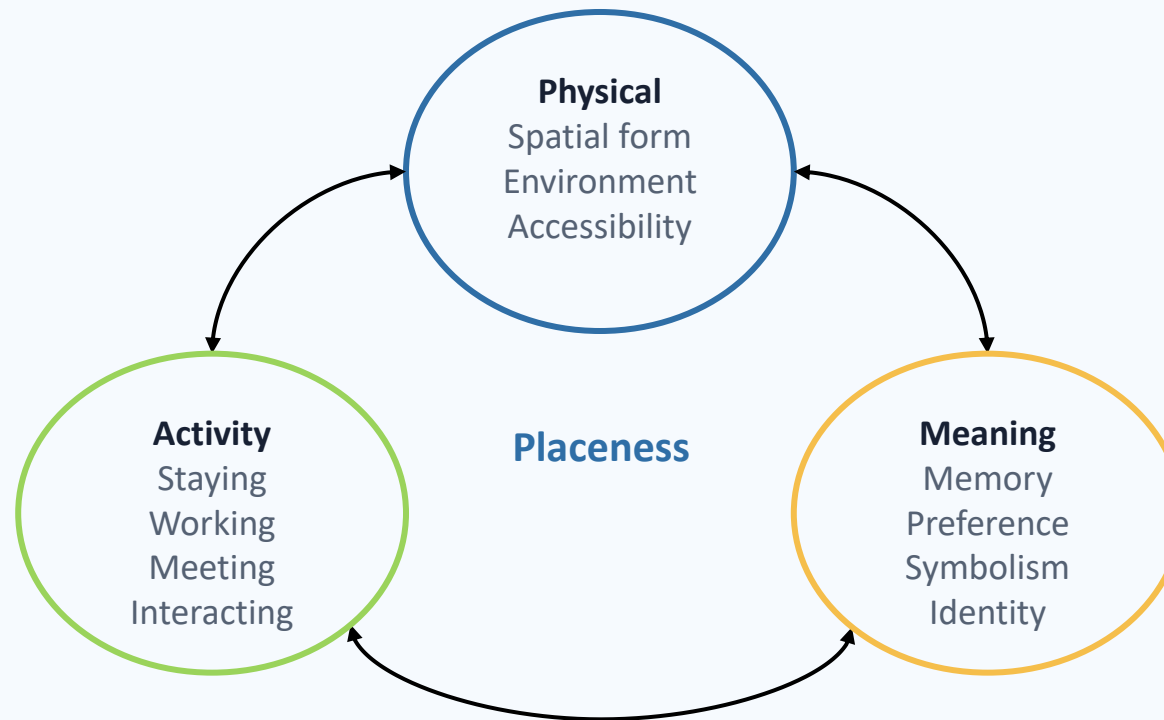
Activity-related characteristics

Behaviors, uses, and social interactions within space

Meaning-related characteristics

Memory, preference, symbolism, and identity formed through experience

Components of Placeness



Placeness emerges through the interaction of these three components.

Relph's Core Components of Placeness

Relph explains the identity of place through the relationship among physical setting, activities, and meanings.

Relph's definition

Physical setting

Material and spatial conditions that form the setting of experience.

Activities

Actions and patterns of use that occur in the place.

Meanings

Values, memories, and interpretations attached to the place.

Reference: Edward Relph, Place and Placelessness (1976)

Placeness Factors in Prior Studies

Placeness factors were selected and similar items were integrated according to the environmental characteristics of commercial spaces.

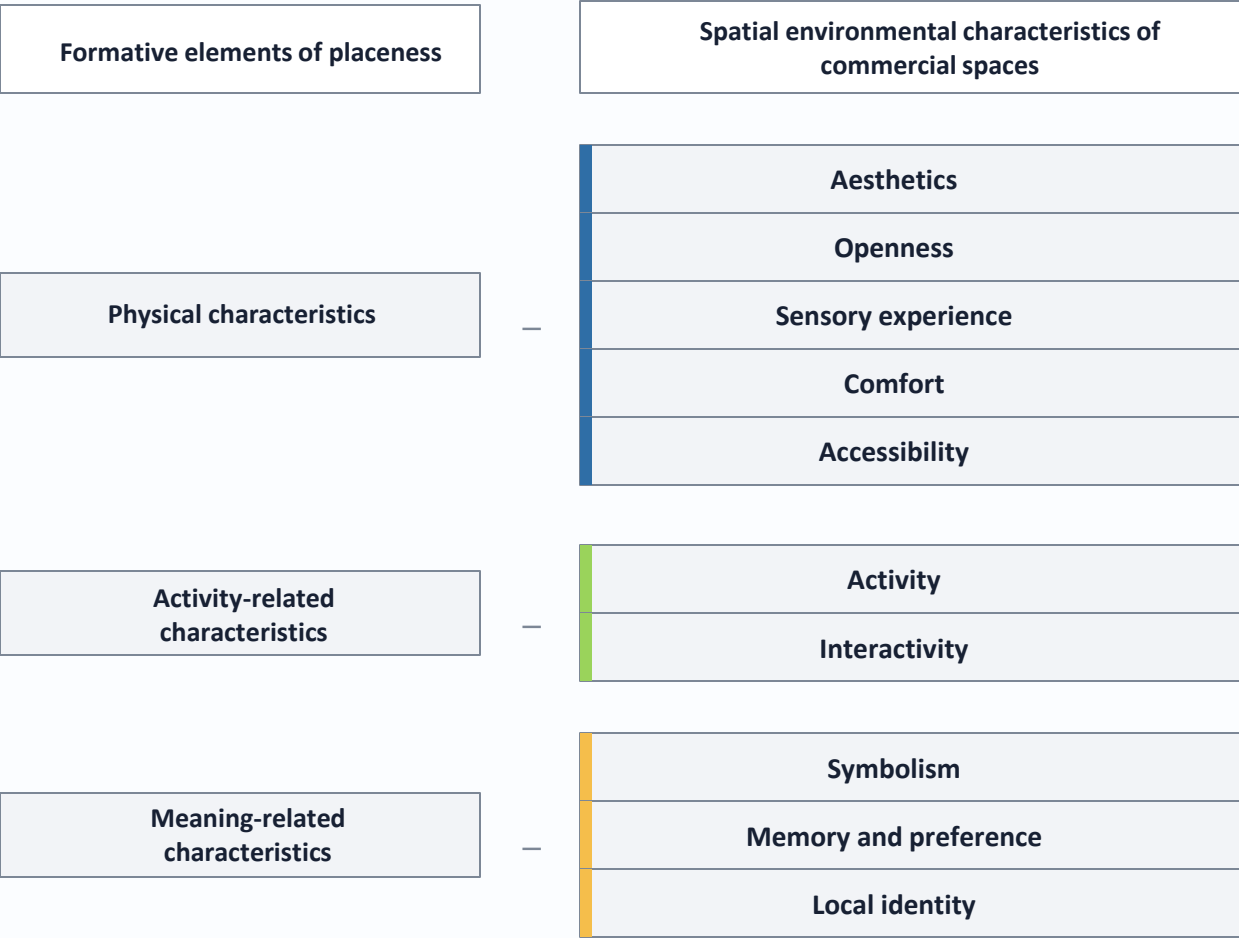
	Author	Title	Target space	Placeness factors
1	Guo, Hwang & Lee (2025)	User perception analysis of museum spatial design based on place characteristics	Museum	[Physical] accessibility, territoriality, attractiveness, comfort [Activity] interactivity, convenience [Meaning] symbolism, educational value
2	Na, Lee & Lee (2025)	Analysis of placeness of major places in Seoul using text mining and large language models	71 major places in Seoul	Symbolism, culturality, everydayness, specificity, vitality, quietness, naturalness
3	Sun & Yu (2021)	Dimension and formation of placeness of commercial public space in city center	Commercial space	[Physical] traffic convenience, visibility, safety, comfort, openness [Activity] activity, seating, environmental pleasure [Meaning] regional landmarkness, functional context
4	Lyu (1996)	A study on embodiment factors of sense of place in interior space	Residential space	Centrality, boundary, control, safety, comfort, symbolism, orientation, sociality, interior design elements
5	Lee & Kim (2025)	Spatial design strategy of large cafés from the perspective of complex cultural space	Eight large cafés	Complexity, accessibility, participation, symbolism, expertise, historicity, experience, diversity, aesthetics, culture, openness, continuity
6	Waxman (2008)	The Coffee Shop: Social and physical factors influencing place attachment	Three cafés in Alabama	[Physical] hygiene, scent, lighting, furniture, exterior view, seating layout [Activity] staying, territoriality, stability, anonymity, productivity, interaction [Meaning] emotional bond



Integrated into 3 dimensions and 10 placeness factors for commercial spaces

Reconstructed Placeness Factors for Commercial Spaces

Detailed placeness factors were reconstructed according to the spatial environmental characteristics of commercial spaces.



Definitions and Keywords of Placeness Factors

Aesthetics	visual quality, design, atmosphere	Activity	work, reading, staying, gathering
Openness	large windows, views, spaciousness	Interaction	staff, companions, social exchange
Sensory experience	light, smell, sound, taste, tactile comfort	Symbolism	landmark, uniqueness, representative image
Accessibility	location, transit, parking, entry	Memory & preference	revisit intention, attachment, favorite place
Comfort	cleanliness, seating, temperature, crowding	Local identity	neighborhood character, local context

Review Expressions by Placeness Factor

Each factor was defined so that it can be identified through expressions that appear in user reviews.

Openness	"The space feels open because the windows are large."
Accessibility	"Parking was inconvenient."
Comfort	"The restroom was clean and well managed."
Activity	"It is good for studying or working."
Memory & preference	"I want to visit this café again."

⋮

Research Target and Scope

The empirical target is cafés in Seoul, selected as a case for testing the review-based placeness quantification approach.

Study area

Seoul, South Korea

Spatial target

Cafés and bakery/café-related commercial spaces

Scope limitation

The result should be interpreted as an empirical case, not a generalization to all space types.

Overview of Research

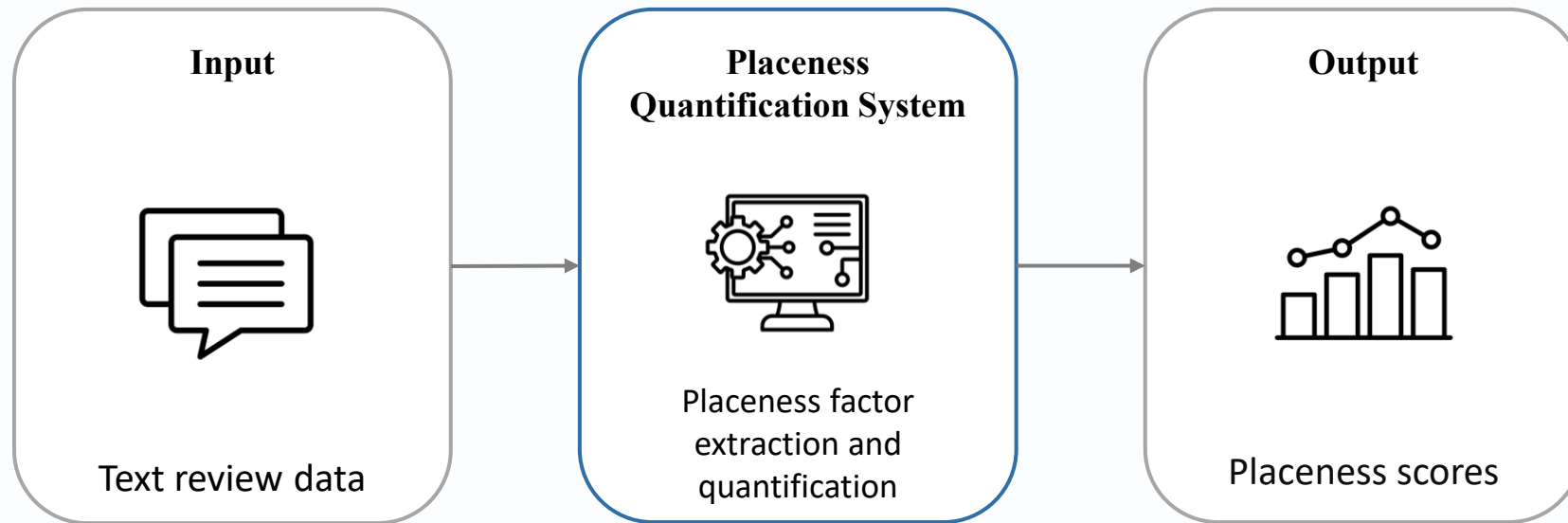
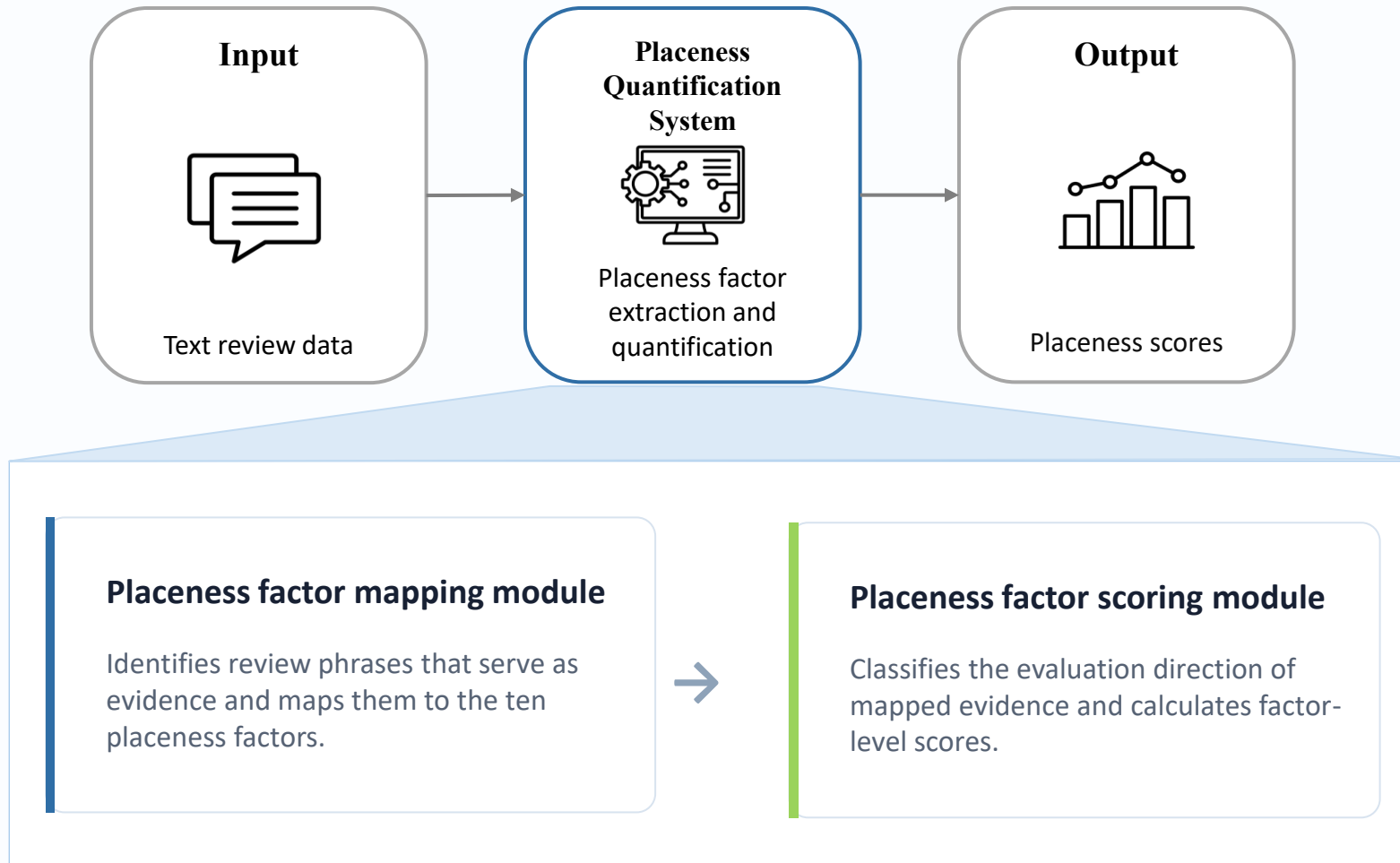


Figure 1. Overview of placeness-based text review quantification approach

Internal Modules of the Quantification System



Placeness Factor Mapping Module

The GPT-4o-mini API is used to map review phrases to placeness factors through an instruction prompt.

Task

Map review phrases to one or more placeness factors.

Definitions

Provide definitions of the ten placeness factors.

Keywords

Provide keywords and review-observable expressions.

Examples

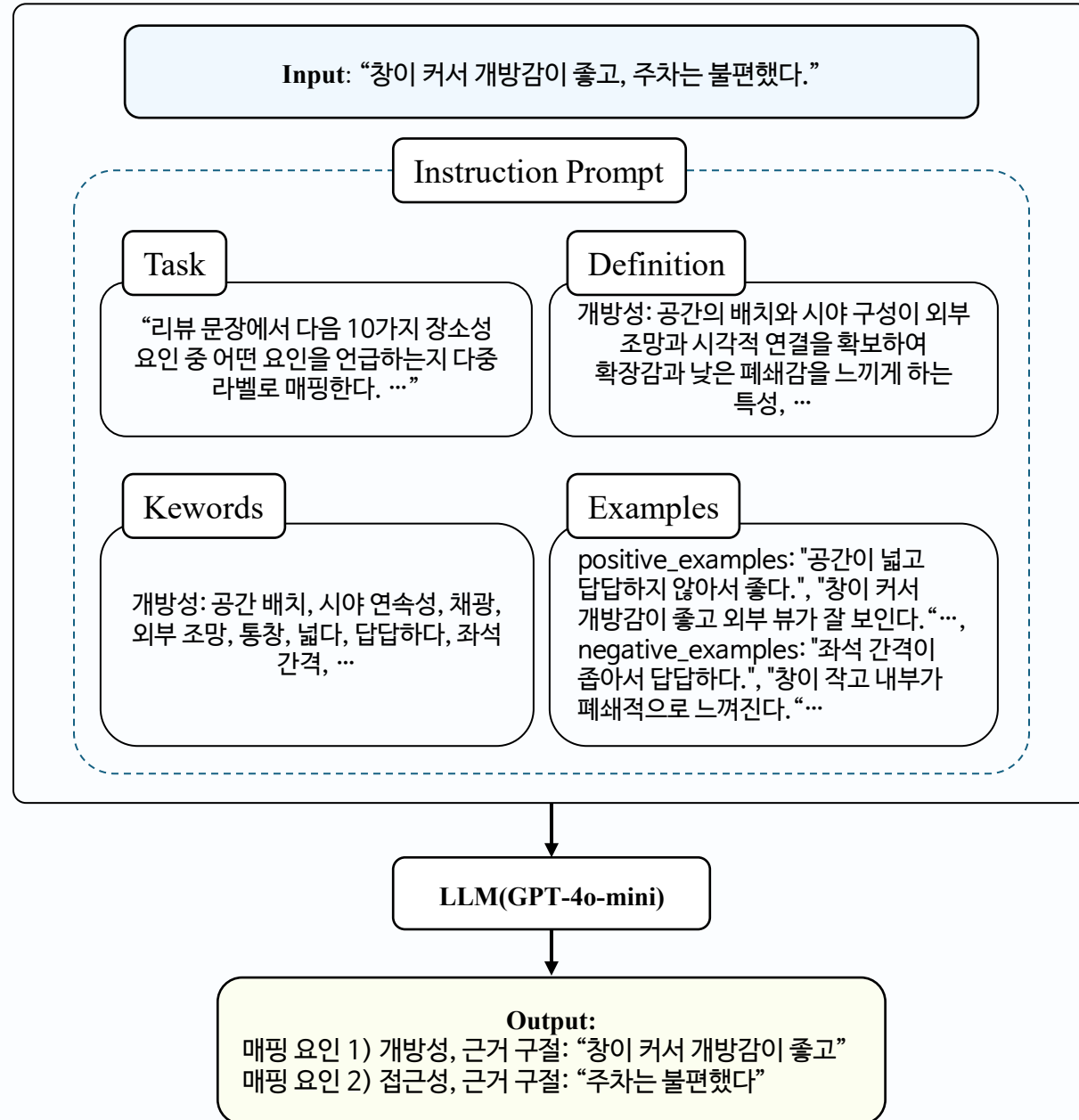
Provide positive and negative examples for each factor.

Output

Return mapped factor and supporting phrase.

Placeness Factor Mapping Module Example

The GPT-4o-mini API is used to map review phrases to placeness factors through an instruction prompt.



Placeness Factor Scoring Module

Phrase-level score

Positive phrase: +1
Neutral or mixed phrase: 0
Negative phrase: -1

Factor score

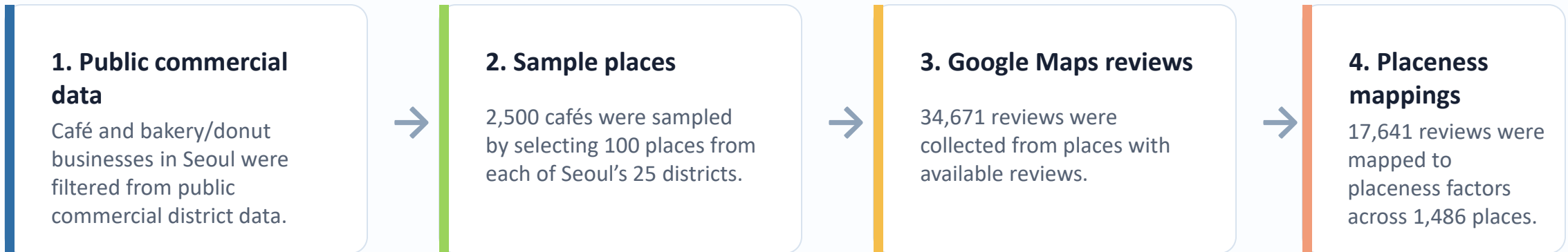
$(\text{positive evidence count} - \text{negative evidence count}) / \text{total evidence count}$
for the factor

Mention share -> Weight

Evidence count for each factor / all placeness evidence for the place

Score range: -1 to +1. Negative values are preserved when negative evidence is dominant.

Data Used in This Study



Data may be updated as review collection expands.

Demo and Application

https://space-agent.streamlit.app/

리뷰

주차공간 꽤 넓습니다. 매장은 1층과 2층으로 구분되어 있고, 신기하게 주문하면 가져다 주시고 치워주시네요. 커피 맛 괜찮았고 화장실도 깔끔하고 관리 잘 되어 있습니다. 전체적인 인테리어는 노출콘트리트~? 인 것 같더라고요. 넓고 쾌적한 합니다. 하지만, 의자가 나무로 되어 있어 조금 불편하구요. 좌식으로 앉는 곳도 생각보다 많이 불편하더라고요. 조명이 어두워서 2층 올라가는 계단 조심하셔야 합니다. 재방문 의사는 있습니다. 끝!

매핑 결과

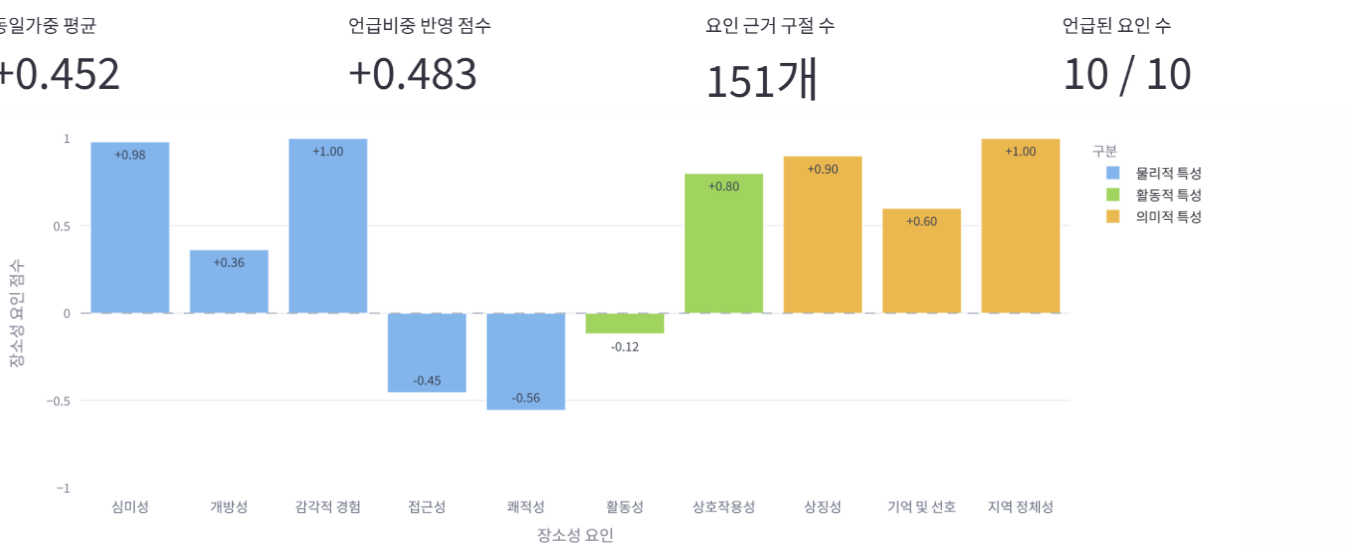
근거 구절	매핑 요인	감성 방향	구절별 점수	매핑 근거
주차공간 꽤 넓습니다.	접근성	긍정	+1	주차 공간에 대한 긍정 언급이므로 접근성 요인
매장은 1층과 2층으로 구분되어 있고, 신기하게 주문하면 가져다 주시고 치워주시네요.	활동성	긍정	+1	공간의 구조와 서비스 방식이 다양한 활동을 지원하
화장실도 깔끔하고 관리 잘 되어 있습니다.	쾌적성	긍정	+1	청결과 관리 상태에 대한 긍정 언급이므로 쾌적성 요
전체적인 인테리어는 노출콘트리트~?	심미성	중립	0	인테리어 디자인에 대한 언급이므로 심미성 요인
넓고 쾌적한 합니다.	개방성	긍정	+1	공간의 넓음에 대한 긍정 언급이므로 개방성 요인
의자가 나무로 되어 있어 조금 불편하구요.	활동성	부정	-1	의자에 대한 불편함 언급이지만 활동성과 관련이 있
좌식으로 앉는 곳도 생각보다 많이 불편하더라고요.	활동성	부정	-1	좌식 공간의 불편함 언급이므로 활동성 요인
조명이 어두워서 2층 올라가는 계단 조심하셔야 합니다.	감각적 경험	부정	-1	조명 상태에 대한 언급이므로 감각적 경험 요인
재방문 의사는 있습니다.	기억 및 선호	긍정	+1	재방문 의사에 대한 긍정 언급이므로 기억 및 선호 요

이 리뷰에서는 총 9개의 장소성 근거 구절이 추출되었습니다. 긍정 긍정 긍정 중립 긍정 부정 부정 부정 긍정

- Shows the process from the placeness factor framework to review mapping results and score calculation.
- Allows users to examine which review phrases support each factor score.

Demo and Application

<https://space-agent.streamlit.app/>



구분	요인	긍정	중립	혼합	부정	근거 수	장소성 요인 점수	언급 비중	가중 점수	근거 예시
물리적 특성	심미성	51	1	0	0	52	+0.981	34.4%	+0.338	그래도 찻기 좋은 카페예요. 마당이 워낙 넓고 예뻐서 포토존 역할도
물리적 특성	개방성	7	1	0	3	11	+0.364	7.3%	+0.026	내부에 좌식 공간도 넉넉해서 한옥 분위기를 느끼며 여유롭게 즐길
물리적 특성	감각적 경험	8	0	0	0	8	+1.000	5.3%	+0.053	분위기는 고요했고, 페이스트리는 여전히 따뜻했으며, 음료는 차갑
물리적 특성	접근성	6	0	0	16	22	-0.455	14.6%	-0.066	안국역 근처에 있는 핫한 카페다 안국역에서 도보로 5분안에 갈 수
물리적 특성	쾌적성	2	0	0	7	9	-0.556	6.0%	-0.033	사람이 많고 에어컨이 안좋습니다. 화장실 아주 더럽고, 테이블은
활동적 특성	활동성	6	3	0	8	17	-0.118	11.3%	-0.013	주말은 사람이 너무 많아 줄서야들어가는 인기좋은 곳. 다양한 좌식
활동적 특성	상호작용성	9	0	0	1	10	+0.800	6.6%	+0.053	매우 다양한 사람들을 두루두루 구경할 수 있다. 직원들은 친절했고
의미적 특성	상징성	9	1	0	0	10	+0.900	6.6%	+0.060	한옥 어니언. 삼단 형태의 한옥이 주는 고유한 매력이 깃든 카페 전
의미적 특성	기억 및 선호	4	0	0	1	5	+0.600	3.3%	+0.020	입도 눈도 즐거운 곳 입니다! 한번쯤 방문 여기가 바로 안국속 천국
의미적 특성	지역 정체성	7	0	0	0	7	+1.000	4.6%	+0.046	전통적인 중부식 '모자' 구조한옥입니다. 외관과 분위기 덕분에 향

- Shows the process from the placeness factor framework to review mapping results and score calculation.
- Allows users to examine which review phrases support each factor score.
- Can be extended to place comparison, spatial search, and personalized recommendation.

Conclusion and Limitations

- This study confirmed that placeness can be quantified using user-generated text review data.
- The proposed scoring framework produces comparable indicators based on placeness factors.
- Review-based quantification can complement fieldwork- and survey-based evaluation methods.
- The system can be extended to place recommendation, spatial search, and place comparison.
- Limitations remain because the empirical target is cafés in Seoul and online reviews do not represent every user experience.

From qualitative place experience to comparable quantitative placeness indicators

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